

Are-new-hires-getting-paid-too-much.R

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```
# Load the packages
```

```
library("dplyr")
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
library("readr")
```

```
library("broom")
```

```
library("ggplot2")
```

```
# Import the data
```

```
pay <- read_csv("~/Desktop/Data/fair_pay_data.csv")
```

```
## Rows: 1470 Columns: 5
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): department, new_hire, job_level
```

```
## dbl (2): employee_id, salary
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# Get an overview of the data
```

```
summary(pay)
```

```
##   employee_id      department      salary      new_hire
##   Min.   : 1.0    Length:1470    Min.    : 43820   Length:1470
##   1st Qu.: 491.2   Class :character  1st Qu.: 59378   Class :character
##   Median :1020.5   Mode  :character  Median : 70425   Mode  :character
##   Mean    :1024.9                Mean    : 74142
```

```
## 3rd Qu.:1555.8          3rd Qu.: 84809
## Max.    :2068.0          Max.    :164073
## job_level
## Length:1470
## Class :character
## Mode  :character
##
##
##
```

```
# Check average salary of new hires and non-new hires
pay %>%
  group_by(new_hire)%>%
  summarize(avg_salary = mean(salary))
```

```
## # A tibble: 2 x 2
##   new_hire avg_salary
##   <chr>      <dbl>
## 1 No          73425.
## 2 Yes         76074.
```

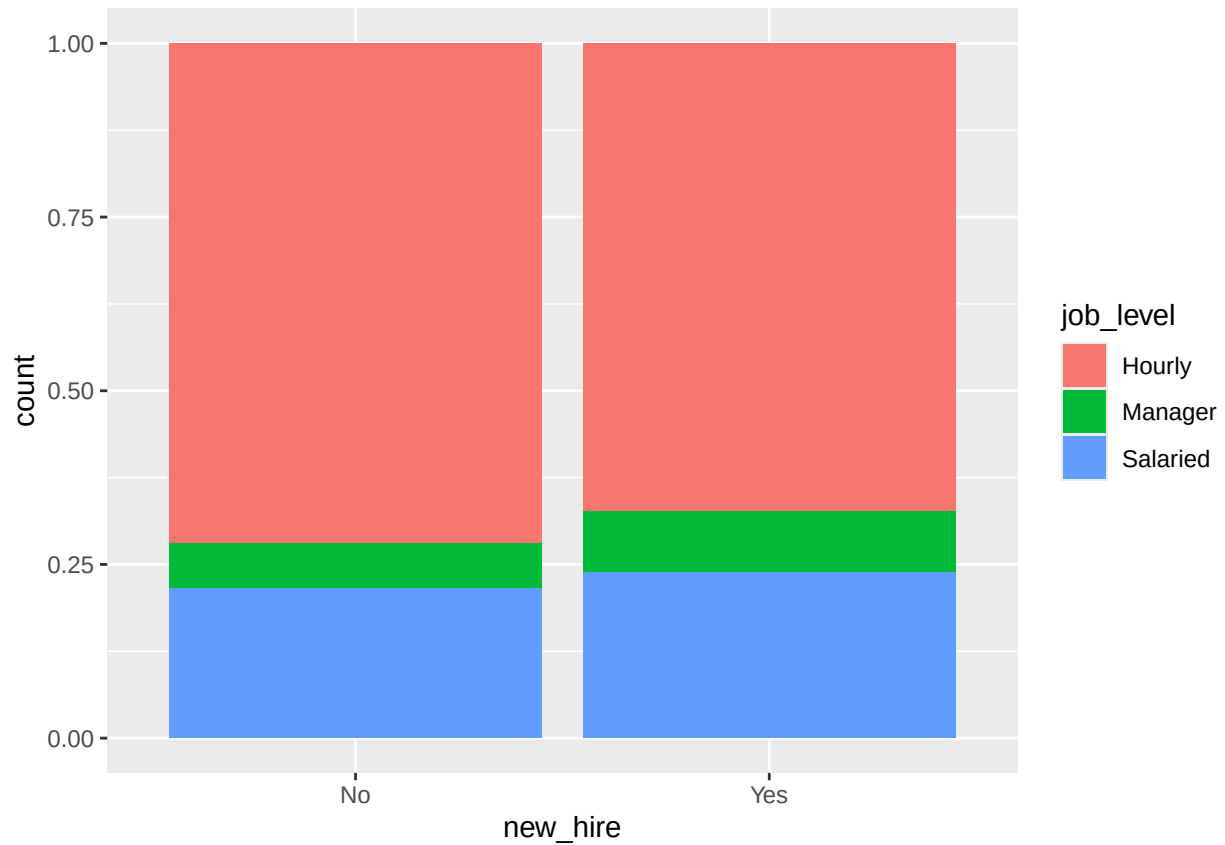
```
# Perform the correct statistical test
t.test(salary ~ new_hire, data = pay)
```

```
##
## Welch Two Sample t-test
##
## data: salary by new_hire
## t = -2.3437, df = 685.16, p-value = 0.01938
## alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
## 95 percent confidence interval:
## -4869.4242 -429.9199
## sample estimates:
## mean in group No mean in group Yes
##      73424.60      76074.28
```

```
# Do the same test, and tidy up the output
t.test(salary ~ new_hire, data = pay) %>%
  tidy()
```

```
## # A tibble: 1 x 10
##   estimate estimate1 estimate2 statistic p.value parameter conf.low conf.high
##   <dbl>    <dbl>    <dbl>    <dbl>  <dbl>    <dbl>    <dbl>    <dbl>
## 1 -2650.    73425.    76074.    -2.34  0.0194    685.    -4869.    -430.
## # i 2 more variables: method <chr>, alternative <chr>
```

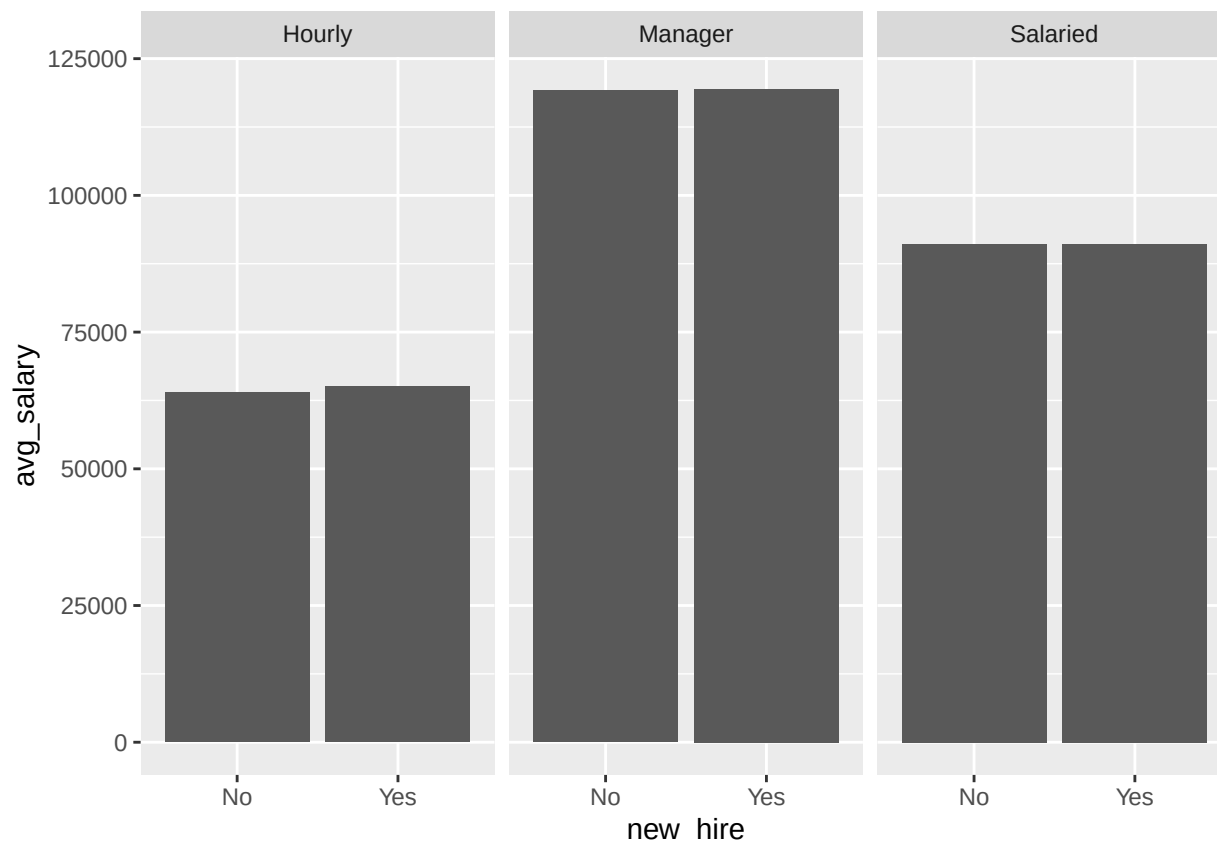
```
# Create a 100% filled stacked bar chart
pay %>% ggplot(aes(x = new_hire, fill = job_level))+ geom_bar(position = "fill")
```



```
# Calculate the average salary for each group of interest
pay_grouped <- pay %>%
  group_by(new_hire, job_level) %>%
  summarize(avg_salary = mean(salary))
```

```
## 'summarise()' has grouped output by 'new_hire'. You can override using the
## '.groups' argument.
```

```
# Graph the results using facet_wrap()
pay_grouped %>%
  ggplot(aes(x=new_hire, y=avg_salary)) + geom_col() + facet_wrap(~job_level)
```



```
# Filter the data to include only hourly employees
pay_filter <- pay %>% filter(job_level == "Hourly")
```

```
# Test the difference in pay
t.test(salary ~ new_hire, data = pay_filter) %>%
  tidy()
```

```
## # A tibble: 1 x 10
##   estimate estimate1 estimate2 statistic p.value parameter conf.low conf.high
##   <dbl>     <dbl>     <dbl>     <dbl>   <dbl>     <dbl>     <dbl>     <dbl>
## 1  -1107.    63966.    65073.    -1.75  0.0807     500.    -2349.     136.
## # i 2 more variables: method <chr>, alternative <chr>
```

```
# Run the simple regression
model_simple <- lm(salary ~ new_hire, data = pay)
```

```
# Display the summary of model_simple
model_simple %>%
  summary()
```

```
##
## Call:
## lm(formula = salary ~ new_hire, data = pay)
##
## Residuals:
```

```
##      Min      1Q Median      3Q      Max
## -32255 -14466  -3681  10740  87998
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  73424.6      577.2  127.200  <2e-16 ***
## new_hireYes  2649.7      1109.4    2.388   0.017 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18900 on 1468 degrees of freedom
## Multiple R-squared:  0.003871, Adjusted R-squared:  0.003193
## F-statistic: 5.705 on 1 and 1468 DF, p-value: 0.01704
```

```
# Display a tidy summary
model_simple %>%
  tidy()
```

```
## # A tibble: 2 x 5
##   term      estimate std.error statistic p.value
##   <chr>      <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)  73425.      577.    127.     0
## 2 new_hireYes  2650.      1109.     2.39  0.0170
```

```
# Run the multiple regression taking Job Level into consideration
model_multiple <- lm(salary ~ new_hire + job_level, data = pay)

# Display the summary of model_multiple
model_multiple %>%
  summary()
```

```
##
## Call:
## lm(formula = salary ~ new_hire + job_level, data = pay)
##
## Residuals:
##      Min      1Q Median      3Q      Max
## -21013  -7091   -425   6771  44322
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    64049.3     308.3  207.722  <2e-16 ***
## new_hireYes      782.7     524.8   1.491   0.136
## job_levelManager 54918.8     915.3  60.001  <2e-16 ***
## job_levelSalaried 26865.6     567.2  47.369  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8930 on 1466 degrees of freedom
## Multiple R-squared:  0.7779, Adjusted R-squared:  0.7775
## F-statistic: 1712 on 3 and 1466 DF, p-value: < 2.2e-16
```

```
# Display a tidy summary
model_multiple %>%
  tidy()
```

```
## # A tibble: 4 x 5
##   term                estimate std.error statistic  p.value
##   <chr>              <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)      64049.     308.    208.      0
## 2 new_hireYes        783.     525.     1.49 1.36e- 1
## 3 job_levelManager  54919.     915.    60.0      0
## 4 job_levelSalaried 26866.     567.    47.4 7.39e-298
```

```
# New Hires are paid about the same as current employees when job level is considered.
```